

Probability vector representation of the Schrödinger equation and Leggett-Garg-type experiments

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Brief summary

We describe the dynamics of generic N-level quantum systems via a probability vector representation of the Schrödinger equation and define a precise notion of no-signaling in time (NSIT) for the probability distributions of non-commuting observables. This provides a systematic way of identifying the interferences responsible for non-classical behavior. In addition, we introduce an interference witness measure to quantify violations of NSIT for arbitrary general probabilistic states.

Main result

Schrödinger equation

$$i\hbar \frac{d}{dt} \hat{\rho}(t) = [\hat{H}, \hat{\rho}(t)]$$

Probability vector representation

$$\frac{d}{dt} p_n(k, t) = \sum_{n'=1}^{N^2-1} \sum_{k'=1}^N H_{nn'}(k, k') p_{n'}(k', t)$$

Key steps in the derivation

Bloch representation: $\hat{\rho}(t) = \frac{1}{N} \left(\hat{I} + \sum_{n=1}^{N^2-1} \langle \lambda_n(t) \rangle \hat{\lambda}_n \right)$ with expectation values $\langle \lambda_n(t) \rangle = \text{Tr} [\hat{\lambda}_n \hat{\rho}(t)]$

Time derivatives: $\frac{d}{dt} \langle \lambda_n(t) \rangle = \text{Tr} \left[\hat{\lambda}_n \frac{d}{dt} \hat{\rho}(t) \right] = \frac{1}{i\hbar} \text{Tr} [\hat{\lambda}_n [\hat{H}, \hat{\rho}(t)]] = \frac{1}{i\hbar} \text{Tr} [\hat{\rho}(t) [\hat{\lambda}_n, \hat{H}]]$

$$\Rightarrow \frac{d}{dt} \langle \lambda_n(t) \rangle = \frac{1}{\hbar} \sum_{n'=1}^{N^2-1} h_{nn'} \langle \lambda_{n'}(t) \rangle \quad (\text{Schrödinger equation in terms of expectation values})$$

$$h_{nn'} := -\frac{i}{N} \text{Tr} [\hat{\lambda}_{n'} [\hat{\lambda}_n, \hat{H}]] = \frac{i}{N} \text{Tr} [\hat{H} [\hat{\lambda}_n, \hat{\lambda}_{n'}]] = -\frac{1}{N} \sum_{n''=1}^{N^2-1} \gamma_{nn''} \text{Tr} [\hat{H} \hat{\lambda}_{n''}]$$

Spectral decomposition: $\hat{\lambda}_n = \sum_{k=1}^N \lambda_n(k) \hat{P}_n(k)$ Emergent probability: $p_n(k, t) = \text{Tr} [\hat{P}_n(k) \hat{\rho}(t)]$
of $\lambda_n(k)$ for the observable $\hat{\lambda}_n$ in the state $\hat{\rho}(t)$

$$\frac{d}{dt} p_n(k, t) = \text{Tr} \left[\hat{P}_n(k) \frac{d}{dt} \hat{\rho}(t) \right] = \frac{1}{i\hbar} \text{Tr} [\hat{P}_n(k) [\hat{H}, \hat{\rho}(t)]] = \frac{1}{i\hbar} \text{Tr} [\hat{\rho}(t) [\hat{P}_n(k), \hat{H}]]$$

$$\text{Use } \frac{1}{i\hbar} [\hat{P}_n(k), \hat{H}] = \sum_{n'=1}^{N^2-1} \sum_{k'=1}^N H_{nn'}(k, k') \hat{P}_{n'}(k') \quad \text{where } H_{nn'}(k, k') := \frac{\lambda_{n'}(k')}{N i \hbar} \text{Tr} [\hat{\lambda}_{n'} [\hat{P}_n(k), \hat{H}]]$$

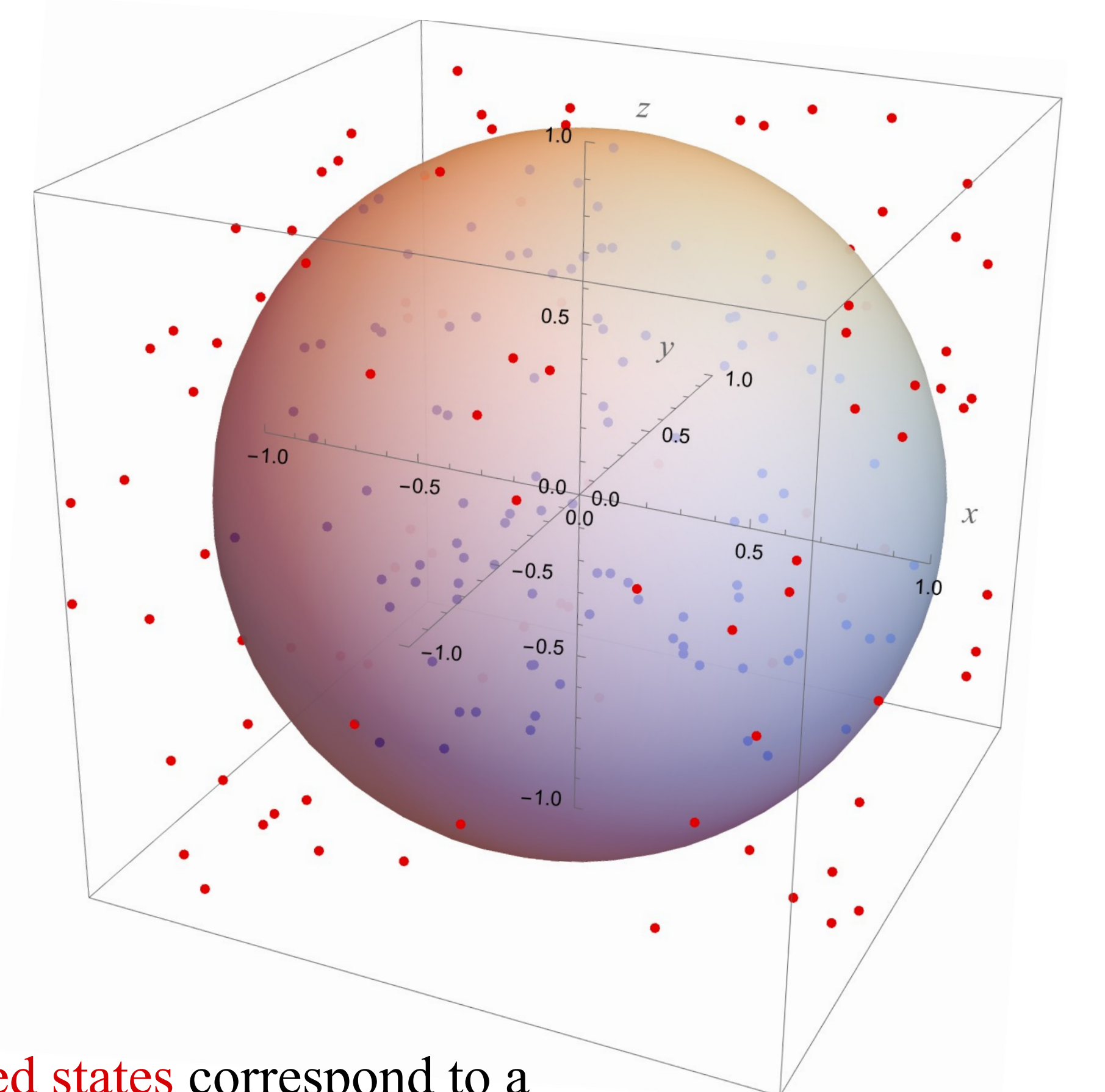
Single qubit state space

Bloch sphere inequality:

$$\langle \sigma_x \rangle^2 + \langle \sigma_y \rangle^2 + \langle \sigma_z \rangle^2 \leq 1$$

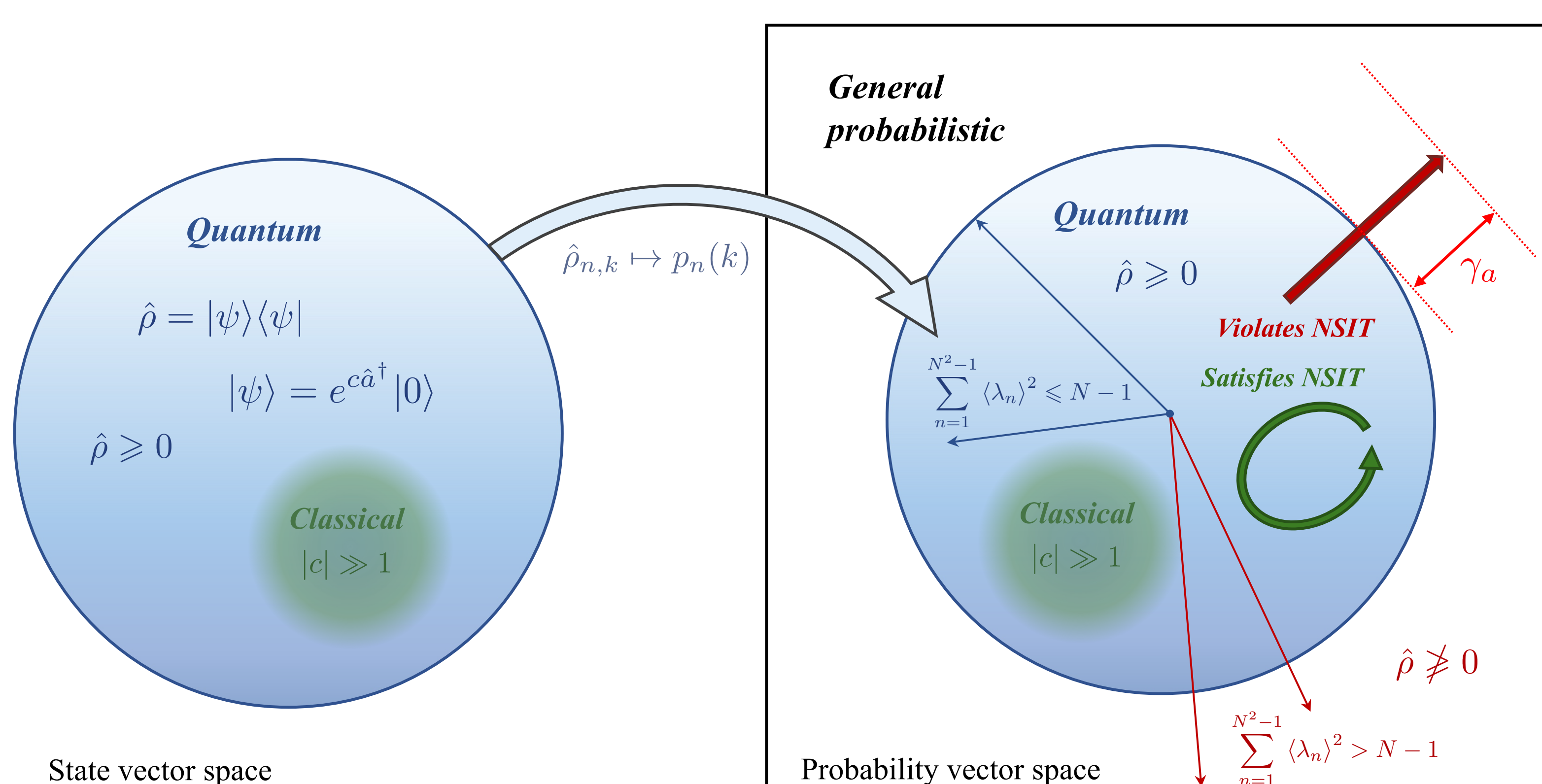
General probabilistic state space:

$$\langle \sigma_a \rangle_{\text{cl}} := (+1) \bar{p}_a(+1, t) + (-1) \bar{p}_a(-1, t)$$



Red states correspond to a probability distribution that does not align with our traditional understanding of quantum mechanics.

Schematic illustration of our approach



Interference witness measure

No-signaling in time (NSIT)

serves as test for noninvasiveness

$$\mathcal{P}(Q, t) - \mathcal{P}(Q, t)_{|(\tilde{Q}, \tilde{t})} = 0$$

Pseudo-density post-measurement matrix:

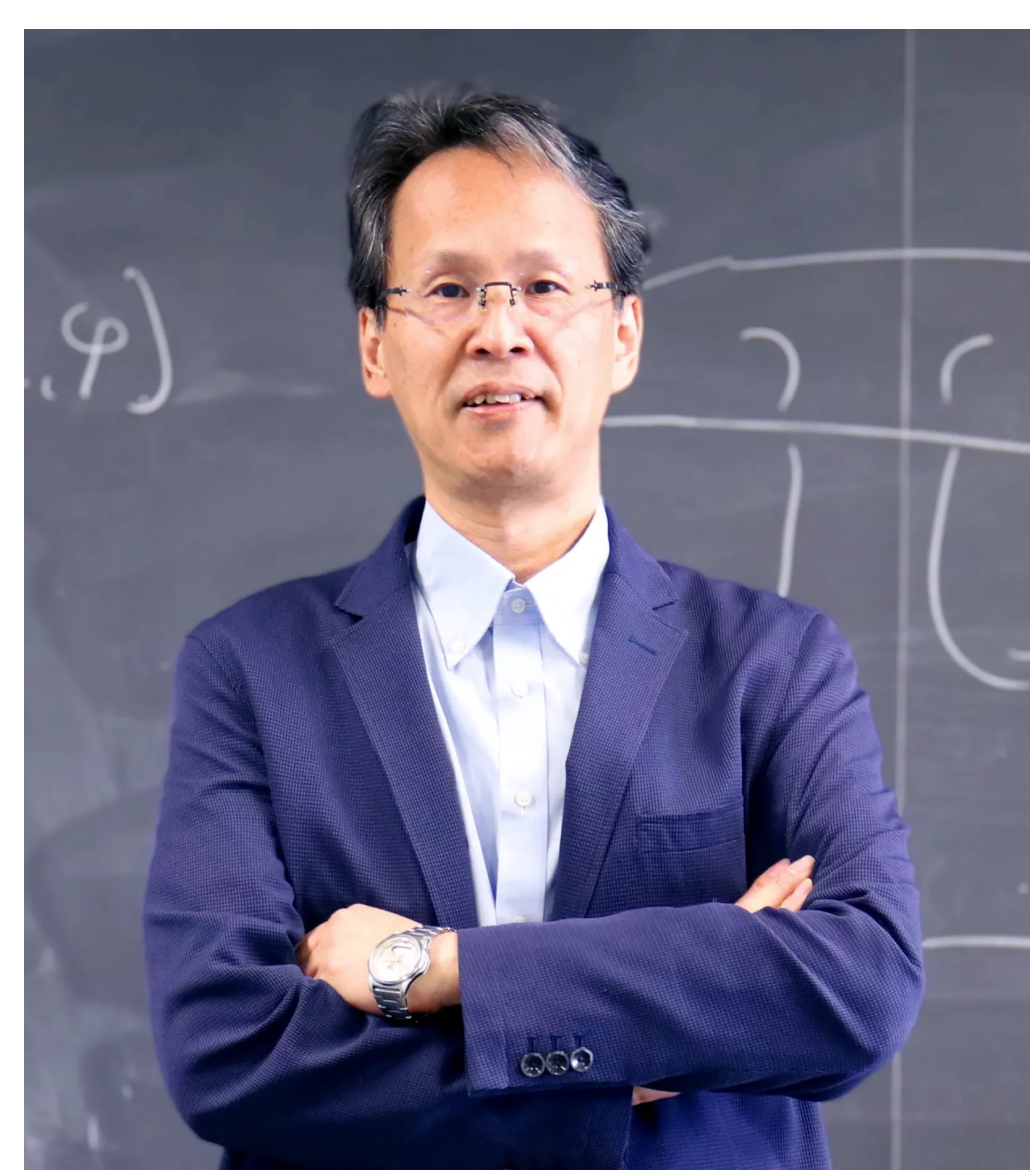
$$\hat{\rho}'_{n,k} = \frac{1}{N} \left(\hat{I} + \sum_{n'=1}^{N^2-1} \langle \lambda_{n'} \rangle'_{n,k} \hat{\lambda}_{n'} \right) = \frac{1}{N} \left(\hat{I} + \lambda_n(k) \hat{\lambda}_n + \sum_{n' \neq n} \text{Tr} [\hat{\lambda}_{n'} \hat{\rho}(0)] \hat{\lambda}_{n'} \right)$$

Spectral decomposition: $\hat{\rho}'_{n,k} = \sum_{m=1}^N p'_m(n, k) \hat{P}'_m(n, k)$

Interference witness measure:

$$\gamma_{n,k} := \sum_{p'_m(n,k) < 0} |p'_m(n, k)|$$

Negative eigenvalues in the spectrum of the post-measurement density operator indicate **NSIT violations!**



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